Regional Bias in Pretrained and Finetuned Vision Models on GeoDE and Dollar Street

TODO: author names

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Abstract

TODO: evidence needed. Write the abstract only after results and claims are validated.

1 Introduction

Object-recognition models are often compared with aggregate accuracy, but aggregate accuracy can hide uneven performance across regions. This is a concern for geographically diverse visual data because the same object category can appear in different environments, styles, materials, backgrounds, and socioeconomic contexts. Prior work on Dollar Street suggests that object-recognition performance may vary across countries and income levels [DeVries et al., 2019]. This motivates evaluating models not only by overall object accuracy but also by their best- and worst-region performance.

This paper studies regional bias in computer vision models on GeoDE and Dollar Street. GeoDE is a geographically diverse evaluation dataset for object recognition [Ramaswamy et al., 2023]. Dollar Street provides household-object imagery intended to represent geographic and socioeconomic visual diversity [Gaviria Rojas et al., 2022]. These datasets make it possible to ask whether strong object recognition performance is accompanied by stable performance across regions.

The main research question is: how do pretrained and finetuned vision models differ in regional bias across GeoDE and Dollar Street? The paper compares nine model names: CLIP, ConvNeXt, DeiT, DINO, MAE, MaxViT, ResNet, SigLIP, and Swin. It evaluates two settings: pretrained ImageNet evaluation and finetuned evaluation. The analysis uses completed result files and derived tables; it does not train new models inside the paper workspace.

The central metrics are object accuracy, best-region accuracy, worst-region accuracy, and regional bias gap. The regional bias gap is the difference between best-region and worst-region accuracy. This metric is simple, but it helps prevent a purely aggregate reading of model performance. A model can have high object accuracy and still show a large regional gap. A model can also have a small regional gap because it performs poorly everywhere. For that reason, this paper interprets object accuracy and regional gap together.

The first draft results suggest three broad patterns. First, finetuning is associated with much higher object accuracy for every model-dataset pair in the derived table. Second, finetuning is also associated with larger regional bias gaps in every model-dataset pair. Third, the dataset matters: finetuned GeoDE results have higher mean object accuracy and lower mean regional gap than finetuned Dollar Street results. These are descriptive findings and should not be read as causal claims.

The contribution of this paper is a reproducible first analysis of regional object-recognition performance across two datasets, nine models, and two evaluation settings. It also provides a workflow for separating empirical claims from literature claims. Empirical claims come from normalized result tables and figures; literature claims come from verified citation entries. TODO: sharpen contributions after the related work and dataset audit are expanded.

2 Related Work

2.1 Geographic Bias in Computer Vision

Prior work has raised concerns that object recognition systems may not perform equally across geographic and socioeconomic contexts. DeVries et al. [2019] evaluate object recognition with Dollar Street imagery and report disparities across countries and income levels. This paper builds on that motivation but uses a different empirical setup: it compares pretrained ImageNet and finetuned evaluations across two datasets and nine model families. The current draft therefore treats prior findings as motivation, not as evidence for the exact mechanisms behind the new results.

GeoDE was introduced as a geographically diverse evaluation dataset for object recognition [Ramaswamy et al., 2023]. It is directly relevant because this paper uses GeoDE as one of the two evaluation datasets and reports region-level accuracy and regional bias gap. Dollar Street provides household-object imagery intended to represent geographic and socioeconomic diversity [Gaviria Rojas et al., 2022]. Together, these datasets support a research design that evaluates model behavior beyond a single aggregate accuracy value.

2.2 Dataset Bias and Domain Shift

Dataset bias has long been recognized as a challenge for visual recognition. Torralba and Efros [2011] show that recognition datasets can contain dataset-specific biases that affect generalization across datasets. Later work on ImageNet generalization and natural distribution shifts similarly suggests that benchmark performance does not fully characterize performance under new data conditions [Recht et al., 2019, Taori et al., 2020]. This literature supports the paper's decision to compare models across datasets and regions rather than relying only on overall object accuracy.

Broader benchmark work also emphasizes the difficulty of real-world distribution shifts. WILDS frames in-the-wild distribution shifts as a recurring challenge across application domains [Koh et al., 2021]. DomainBed emphasizes careful evaluation design and model-selection choices in domain generalization research [Gulrajani and Lopez-Paz, 2021]. These papers do not answer this paper's regional-bias question directly, but they provide useful methodological caution: descriptive performance gaps should not be overinterpreted as causal mechanisms.

Visual dataset auditing work is also relevant. REVISE provides tools for measuring and mitigating object-, person-, and geography-related biases in visual datasets [Wang et al., 2022]. This supports the broader idea that geography can be an important axis in visual dataset analysis. TODO: evidence needed to connect REVISE-style dataset auditing more directly to the exact regional metrics used here.

2.3 GeoDE and Dollar Street

GeoDE and Dollar Street are central to this paper because they provide regionally meaningful evaluation settings. GeoDE is cited here as the dataset source for geographically diverse object recognition [Ramaswamy et al., 2023]. Dollar Street is cited as the dataset source for household-object imagery designed around geographic and socioeconomic diversity [Gaviria Rojas et al., 2022].

Prior Dollar Street object-recognition work provides direct motivation for asking whether models behave unevenly across regions [DeVries et al., 2019].

The present draft does not reproduce dataset statistics from the dataset papers because the allowed evidence sources for this draft do not yet include a verified local dataset-statistics table. TODO: evidence needed for local split sizes, region counts, and class counts. TODO: evidence needed for how the local object labels map to the official dataset descriptions.

2.4 Vision-Language Model Bias and Robustness

CLIP introduced contrastive language-image pretraining and showed that natural language supervision can produce transferable visual models [Radford et al., 2021]. SigLIP modifies language-image pretraining by using a sigmoid loss over image-text pairs [Zhai et al., 2023]. These papers support the model descriptions for the vision-language models used in this paper.

For robustness, Fang et al. [2022] argue that data distribution is important for distributional robustness in CLIP-like training. This is relevant to the discussion because this paper observes that CLIP and SigLIP do not have identical regional-gap behavior. However, the current experiments do not test pretraining data composition directly. The paper should therefore avoid claims that language-image pretraining causes a particular regional robustness pattern.

2.5 Self-Supervised Model Robustness

DINOv2 and MAE provide the main self-supervised model context for this draft. DINOv2 is described as learning robust visual features without supervision [Oquab et al., 2024], while MAE learns visual representations through masked image reconstruction [He et al., 2022]. In this paper's derived results, MAE has the lowest finetuned regional gap on both datasets, but it does not have the highest object accuracy. This makes self-supervised models important for the accuracy-gap trade-off discussion. The draft should still avoid saying that self-supervision causes lower regional bias, because the evidence is descriptive.

2.6 CNN and Transformer Backbones

The evaluated CNN and transformer models are grounded in standard architecture papers. ResNet introduced residual learning for deep convolutional networks [He et al., 2016], and ConvNeXt revisited ConvNet design choices for modern vision architectures [Liu et al., 2022]. DeiT studied data-efficient training and distillation for vision transformers [Touvron et al., 2021]; Swin introduced shifted-window hierarchical attention [Liu et al., 2021]; and MaxViT introduced a multi-axis attention design [Tu et al., 2022]. These citations are used only to describe model families, not to claim anything about their regional bias behavior in this paper.

2.7 Regional Fairness Metrics and Worst-Group Evaluation

This paper uses worst-region accuracy and regional bias gap as descriptive metrics. The motivation is related to worst-group evaluation, where average accuracy can hide weak performance for specific groups [Sagawa et al., 2020]. ObjectNet and related robustness benchmarks also support evaluation beyond conventional test-set accuracy [Barbu et al., 2019]. The regional bias gap used here is simpler than a full fairness analysis, but it helps identify whether high object accuracy is accompanied by uneven regional performance.

3 Problem Definition

The paper studies regional variation in object-recognition performance. Let d denote a dataset, m denote a model, s denote an evaluation setting, and r denote a region. The two datasets are Dollar Street and GeoDE, and the two settings are pretrained ImageNet evaluation and finetuned evaluation. The analysis is based on completed result files rather than new interventions.

For each dataset-model-setting combination, the primary aggregate metric is object accuracy, recorded as `object_acc`. This is the fraction of test images for which the predicted object label matches the ground-truth object label. The regional version of this metric is recorded in the source results as per-region object accuracy and summarized in the normalized table through `best_region`, `worst_region`, `best_region_acc`, and `worst_region_acc`.

The best region is the region with the highest regional object accuracy for a given dataset-model-setting row. The worst region is the region with the lowest regional object accuracy for that same row. The regional bias gap is defined as:

$$\text{region\_bias\_gap}_{d,m,s} = \text{best\_region\_acc}_{d,m,s} - \text{worst\_region\_acc}_{d,m,s}.$$

A larger value indicates a larger difference between the highest-performing and lowest-performing regions. A smaller value indicates more similar best- and worst-region performance, but it does not by itself imply that the model is useful or robust. For example, a model with uniformly low accuracy can have a small regional gap.

This definition follows the general motivation of worst-group evaluation: aggregate accuracy can hide weak performance for particular groups [Sagawa et al., 2020]. The paper uses the term "regional bias gap" descriptively. It does not claim that the gap identifies discrimination, causal mechanisms, or deployment harm. Those questions would require additional evidence about dataset construction, region definitions, user populations, and model use cases. TODO: citation needed for a region-specific fairness metric if the final paper uses stronger fairness language.

The main research question is: how do pretrained and finetuned vision models differ in regional bias across GeoDE and Dollar Street? The secondary questions ask which models have the highest object accuracy, which have the lowest regional gap, whether finetuning is associated with higher object accuracy, whether it is associated with lower or higher regional gap, whether model families differ descriptively, and whether high object accuracy is always accompanied by lower regional bias.

4 Datasets

4.1 GeoDE

GeoDE is used as one of the two regional evaluation datasets. The dataset is cited as a geographically diverse evaluation dataset for object recognition [Ramaswamy et al., 2023]. In the current derived results, GeoDE regions appear with names such as Africa, Americas, EastAsia, Europe, SouthEastAsia, and WestAsia. The paper uses the region labels as they appear in the external result JSON files and the normalized metric table.

The GeoDE results in this draft show high finetuned object accuracy and comparatively small finetuned regional gaps. The dataset-level summary reports mean finetuned object accuracy of 0.9508 and mean finetuned regional bias gap of 0.0234. These numbers are empirical results from the derived tables, not general properties of the dataset.

TODO: evidence needed for local GeoDE split sizes, local class count, image count by region, and any preprocessing differences from the official dataset. TODO: evidence needed for whether all models saw exactly the same local train, validation, and test splits.

4.2 Dollar Street

Dollar Street is used as the second regional evaluation dataset. It provides household-object imagery intended to capture geographic and socioeconomic diversity [Gaviria Rojas et al., 2022]. Prior work using Dollar Street suggests that object-recognition systems may perform unevenly across countries and income levels [DeVries et al., 2019]. This makes Dollar Street directly relevant for studying regional performance gaps.

In the current derived results, Dollar Street regions appear with abbreviated labels such as `af`, `am`, `as`, and `eu`. The paper keeps these labels as reported in the result files. The dataset-level summary reports mean finetuned object accuracy of 0.7211 and mean finetuned regional bias gap of 0.0889. Again, these are results from the evaluated runs, not claims about all possible Dollar Street evaluations.

TODO: evidence needed for the exact meaning of each abbreviated Dollar Street region label in the local split. TODO: evidence needed for local Dollar Street split sizes, class count, image count by region, and object-category distribution.

4.3 Comparison Role in This Paper

The two datasets play complementary roles. GeoDE is directly framed around geographically diverse object recognition, while Dollar Street is framed around geographic and socioeconomic visual diversity. The results suggest that the same models can show different accuracy-gap behavior across the two datasets. This supports the paper's decision to compare both rather than treating one dataset as sufficient evidence for regional robustness.

The draft avoids dataset-statistics claims until the local data audit is complete. In the final paper, this section should include a verified table with dataset split sizes, region labels, object classes, and any filtering or preprocessing choices. TODO: create and cite a local dataset-audit table.

5 Models

The paper evaluates nine model names in the derived result table: CLIP, ConvNeXt, DeiT, DINO, MAE, MaxViT, ResNet, SigLIP, and Swin. These models are grouped into four rough families for discussion: CNNs, transformers, self-supervised models, and vision-language models. The grouping is descriptive and should not be treated as causal evidence.

The CNN group contains ResNet and ConvNeXt. ResNet introduced residual learning for deep convolutional image-recognition networks [He et al., 2016]. ConvNeXt revisited convolutional architecture design in the context of modern vision systems [Liu et al., 2022]. In the current results, ConvNeXt is the highest-accuracy finetuned model on both datasets, while ResNet has the lowest finetuned object accuracy on both datasets. This shows that the CNN grouping contains models with very different performance in the present experiments.

The transformer group contains DeiT, Swin, and MaxViT. DeiT is associated with data-efficient image transformer training and distillation [Touvron et al., 2021]. Swin uses hierarchical shifted-window attention [Liu et al., 2021]. MaxViT uses a multi-axis attention design [Tu et al., 2022]. In the derived results, the transformer models are generally strong in finetuned object accuracy, but their regional gaps vary by dataset and model.

The self-supervised group contains DINO and MAE. DINOv2 is cited here as a self-supervised visual representation model [Oquab et al., 2024], and MAE is cited as a masked image modeling approach [He et al., 2022]. In this paper's results, MAE is notable because it has the lowest finetuned regional gap on both datasets, although it does not have the highest object accuracy.

The vision-language group contains CLIP and SigLIP. CLIP uses natural-language supervision over image-text pairs [Radford et al., 2021], and SigLIP modifies language-image pretraining with a sigmoid loss [Zhai et al., 2023]. Prior work suggests that CLIP-like robustness can be associated with the distribution of pretraining data [Fang et al., 2022]. The current draft uses that literature as context only; it does not claim to identify the pretraining causes of CLIP or SigLIP regional behavior.

TODO: verify the exact local model identifiers for every model from all `used_config.yaml` files. One inspected local configuration entry for Dollar Street Swin should be rechecked against the intended model naming before the final paper. TODO: add a model-details table listing architecture key, pretrained source, image size, and evaluation setting.

6 Methodology

This paper uses an evidence-driven aggregation workflow rather than a new training procedure. The empirical unit of analysis is a dataset-model-setting result. Each row corresponds to one dataset, one model, and one evaluation setting. The two datasets are Dollar Street and GeoDE, and the nine model names used in the derived table are `clip`, `convnext`, `deit`, `dino`, `mae`, `maxvit`, `resnet`, `siglip`, and `swin`. The two evaluation settings are recorded with the variable `finetune`: `yes` for finetuned evaluation and `no` for pretrained ImageNet evaluation.

The main source table is `derived/processed/all_metrics.csv`. It is produced by reading finetuned results from the external path pattern `results/{dataset}/{model}/test_results.json` and pretrained ImageNet results from `results/imagenet/{dataset}_{model}.json`. The aggregation script writes both CSV and JSON outputs inside the paper workspace and does not modify the external result files. This separation is important because the paper workspace is intended for reproducible analysis and writing, not model training.

The normalized table contains the following columns: `dataset`, `model`, `loss`, `object_acc`, `best_region`, `worst_region`, `best_region_acc`, `worst_region_acc`, `region_bias_gap`, and `finetune`. When `region_bias_gap` is present in a source JSON file, the aggregation keeps that value. When it is absent but both `best_region_acc` and `worst_region_acc` are available, the gap is computed as:

$$\texttt{region\_bias\_gap} = \texttt{best\_region\_acc} - \texttt{worst\_region\_acc}.$$

The validation script checks that this relationship holds in the normalized table.

The finetuning configuration files indicate a common evaluation pipeline with image size 224, batch size 32, Adam optimization, cosine scheduling, label smoothing of 0.1, automatic mixed precision enabled, and test-after-train enabled. The external `used_config.yaml` files checked for this draft report 10 epochs for the inspected dataset-model runs. The training script computes object accuracy as the fraction of correct object predictions, collects per-region correct and total counts when region labels are available, and then reports the best and worst regions by regional object accuracy. TODO: evidence needed to confirm that every model run used exactly the same hyperparameters beyond the inspected `used_config.yaml` files.

The pretrained ImageNet setting is treated as the evaluation setting with `finetune=no` in the derived table. The draft does not claim that this setting is a pure zero-shot benchmark for all architectures, because classifier-head behavior and class-mapping details still need to be documented from an allowed audit artifact. TODO: evidence needed to fully describe the pretrained classifier-head setup for each model.

All analysis is descriptive. The paper compares object accuracy, worst-region accuracy, best-region accuracy, and regional bias gap across models, datasets, and settings. These comparisons can suggest associations, such as whether finetuning is associated with higher object accuracy or a larger regional gap, but they cannot establish causality. The cautious framing is consistent with the broader domain generalization and distribution-shift literature, which emphasizes careful evaluation protocols and limited transfer from one benchmark setting to another [Gulrajani and Lopez-Paz, 2021, Koh et al., 2021].

7 Experiments

The experiments analyzed in this paper consist of already-computed external result files. No new model training is performed inside the paper workspace. Finetuned metrics come from each model's `test_results.json` file under the dataset-specific result directory. Pretrained ImageNet metrics come from JSON files under `results/imagenet`. The paper workspace reads those files, normalizes the metrics, validates the resulting table, and creates derived tables and figures.

The normalized table contains 36 rows: 2 datasets, 9 models, and 2 evaluation settings. The validation step checks that each expected dataset-model-setting combination is present, that `finetune` is only `yes` or `no`, and that the regional bias gap is numerically consistent with the difference between the best-region and worst-region accuracies. The validation also checks that accidental unsupported evaluation columns are not present.

The table-generation scripts produce paper-ready CSV and LaTeX tables. The main tables used in the current draft are the overall metrics table, the pretrained-versus-finetuned comparison table, the regional-gap ranking table, the best/worst region table, the dataset-level summary table, and the model-level summary table. The figure script produces five diagnostic plots: object accuracy by model and dataset, regional bias gap by model and dataset, pretrained-versus-finetuned object accuracy, pretrained-versus-finetuned regional bias gap, and object accuracy versus regional bias gap.

Table 1: Dataset-level summary

dataset	finetune	mean object acc	mean worst region acc	mean best region acc	mean region bias gap	n
dollarstreet	no	0.0192	0.0148	0.0243	0.0095	9.0000
dollarstreet	yes	0.7211	0.6692	0.7580	0.0889	9.0000
geode	no	0.0266	0.0206	0.0321	0.0115	9.0000
geode	yes	0.9508	0.9396	0.9630	0.0234	9.0000

The dataset-level summary in Table 1 is useful for checking the broad scale of the results before interpreting individual models. In the finetuned setting, mean object accuracy is 0.7211 on Dollar Street and 0.9508 on GeoDE. In the pretrained setting, mean object accuracy is much lower: 0.0192 on Dollar Street and 0.0266 on GeoDE. The regional bias gap shows a different pattern: in the finetuned setting, the mean gap is 0.0889 on Dollar Street and 0.0234 on GeoDE; in the pretrained setting, the mean gap is 0.0095 on Dollar Street and 0.0115 on GeoDE.

Table 2 summarizes model-level averages across the two datasets for each evaluation setting. These averages are descriptive only. They are useful for orienting the reader, but the main claims should still be checked against dataset-specific rows because the Dollar Street and GeoDE patterns differ substantially. TODO: add a family-level summary table for CNN, transformer, self-supervised, and vision-language groups.

Table 2: Model-level summary

model	finetune	mean object acc	mean worst region acc	mean best region acc	mean region bias gap	n
clip	no	0.0299	0.0226	0.0337	0.0111	2.0000
clip	yes	0.8421	0.8077	0.8733	0.0657	2.0000
convnext	no	0.0199	0.0143	0.0260	0.0117	2.0000
convnext	yes	0.8664	0.8386	0.8925	0.0538	2.0000
deit	no	0.0184	0.0153	0.0201	0.0048	2.0000
deit	yes	0.8570	0.8243	0.8788	0.0546	2.0000
dino	no	0.0158	0.0099	0.0208	0.0109	2.0000
dino	yes	0.8467	0.8186	0.8691	0.0505	2.0000
mae	no	0.0294	0.0238	0.0365	0.0127	2.0000
mae	yes	0.7980	0.7713	0.8151	0.0437	2.0000
maxvit	no	0.0270	0.0195	0.0382	0.0187	2.0000
maxvit	yes	0.8426	0.8047	0.8694	0.0647	2.0000
resnet	no	0.0254	0.0206	0.0307	0.0101	2.0000
resnet	yes	0.7486	0.7054	0.7797	0.0743	2.0000
siglip	no	0.0208	0.0165	0.0247	0.0082	2.0000
siglip	yes	0.8606	0.8327	0.8837	0.0511	2.0000
swin	no	0.0194	0.0166	0.0231	0.0065	2.0000
swin	yes	0.8617	0.8360	0.8828	0.0468	2.0000

8 Results

8.1 Object Accuracy

The clearest result is that finetuned evaluation has much higher object accuracy than pretrained ImageNet evaluation for every dataset-model pair in the derived table. On Dollar Street, the object-accuracy delta from pretrained to finetuned ranges from 0.5662 for ResNet to 0.7507 for ConvNeXt, as shown in Table 4. On GeoDE, the corresponding delta ranges from 0.8803 for ResNet to 0.9451 for DeiT. This suggests that finetuning is strongly associated with improved object accuracy in the available runs. However, this cannot establish causality because the paper only observes completed evaluations and does not manipulate training conditions.

Within the finetuned setting, GeoDE has higher object accuracy than Dollar Street for all nine models. The highest finetuned GeoDE accuracy is 0.9661 for ConvNeXt, followed by 0.9628 for DeiT and 0.9625 for Swin. The lowest finetuned GeoDE accuracy is 0.9086 for ResNet. On Dollar Street, the highest finetuned accuracy is also ConvNeXt at 0.7667, followed by SigLIP at 0.7621 and Swin at 0.7609. The lowest finetuned Dollar Street accuracy is ResNet at 0.5887.

The pretrained ImageNet setting has low object accuracy across both datasets. On Dollar Street, pretrained object accuracy ranges from 0.0107 for DINO to 0.0288 for MaxViT. On GeoDE, pretrained object accuracy ranges from 0.0177 for DeiT to 0.0451 for CLIP. These low values mean that regional gaps in the pretrained setting should be interpreted carefully. A small gap can reflect uniformly weak performance, not necessarily stable regional robustness.

8.2 Best and Worst Regions

The best-region and worst-region patterns differ sharply by dataset. In the finetuned Dollar Street rows, every model has `eu` as the best region and `af` as the worst region. The best-region accuracy ranges from 0.6356 for ResNet to 0.8061 for ConvNeXt, while the worst-region accuracy ranges from 0.5136 for ResNet to 0.7214 for ConvNeXt. This pattern suggests a consistent regional ordering in the available finetuned Dollar Street results, but the draft cannot explain the mechanism without additional evidence about dataset composition, object categories, and region distributions.

Table 3: Overall metrics by dataset, model, and evaluation setting

dataset	model	finetune	loss	object acc	best region	worst region	best region acc	worst region acc	region bias gap
dollarstreet	clip	no	4.4294	0.0146	eu	af	0.0175	0.0118	0.0057
dollarstreet	clip	yes	1.5627	0.7305	eu	af	0.7784	0.6777	0.1007
dollarstreet	convnext	no	4.2868	0.0160	am	af	0.0234	0.0106	0.0127
dollarstreet	convnext	yes	1.4279	0.7667	eu	af	0.8061	0.7214	0.0848
dollarstreet	deit	no	4.4183	0.0190	as	am	0.0208	0.0152	0.0057
dollarstreet	deit	yes	1.5178	0.7512	eu	af	0.7857	0.6966	0.0891
dollarstreet	dino	no	4.6843	0.0107	af	eu	0.0153	0.0044	0.0110
dollarstreet	dino	yes	1.5633	0.7340	eu	af	0.7668	0.6895	0.0773
dollarstreet	mae	no	4.1876	0.0260	am	eu	0.0292	0.0219	0.0073
dollarstreet	mae	yes	1.7323	0.6674	eu	af	0.6910	0.6222	0.0688
dollarstreet	maxvit	no	4.1550	0.0288	eu	as	0.0452	0.0198	0.0254
dollarstreet	maxvit	yes	1.5243	0.7286	eu	af	0.7726	0.6623	0.1103
dollarstreet	resnet	no	4.1654	0.0225	eu	am	0.0262	0.0199	0.0064
dollarstreet	resnet	yes	2.0059	0.5887	eu	af	0.6356	0.5136	0.1220
dollarstreet	siglip	no	4.2136	0.0165	as	eu	0.0182	0.0131	0.0051
dollarstreet	siglip	yes	1.4584	0.7621	eu	af	0.7945	0.7202	0.0743
dollarstreet	swin	no	4.2285	0.0186	af	as	0.0224	0.0162	0.0063
dollarstreet	swin	yes	1.4694	0.7609	eu	af	0.7915	0.7190	0.0725
geode	clip	no	3.8468	0.0451	EastAsia	WestAsia	0.0498	0.0333	0.0165
geode	clip	yes	0.8170	0.9538	WestAsia	SouthEastAsia	0.9683	0.9377	0.0306
geode	convnext	no	3.8047	0.0239	Americas	EastAsia	0.0287	0.0180	0.0107
geode	convnext	yes	0.7821	0.9661	WestAsia	SouthEastAsia	0.9788	0.9559	0.0229
geode	deit	no	4.0184	0.0177	SouthEastAsia	Americas	0.0193	0.0154	0.0039
geode	deit	yes	0.7943	0.9628	WestAsia	SouthEastAsia	0.9720	0.9520	0.0200
geode	dino	no	4.0929	0.0210	SouthEastAsia	WestAsia	0.0262	0.0153	0.0108
geode	dino	yes	0.7951	0.9595	WestAsia	SouthEastAsia	0.9714	0.9477	0.0238
geode	mae	no	3.7293	0.0329	SouthEastAsia	Africa	0.0437	0.0257	0.0180
geode	mae	yes	0.8839	0.9286	WestAsia	SouthEastAsia	0.9392	0.9205	0.0187
geode	maxvit	no	3.6987	0.0253	WestAsia	EastAsia	0.0312	0.0193	0.0119
geode	maxvit	yes	0.8160	0.9565	WestAsia	SouthEastAsia	0.9662	0.9470	0.0192
geode	resnet	no	3.6951	0.0282	Americas	Africa	0.0351	0.0213	0.0138
geode	resnet	yes	0.9538	0.9086	Europe	SouthEastAsia	0.9238	0.8972	0.0266
geode	siglip	no	3.7253	0.0252	WestAsia	Europe	0.0312	0.0199	0.0113
geode	siglip	yes	0.7972	0.9591	WestAsia	SouthEastAsia	0.9730	0.9452	0.0278
geode	swin	no	3.8006	0.0203	Europe	Africa	0.0238	0.0171	0.0067
geode	swin	yes	0.7906	0.9625	WestAsia	SouthEastAsia	0.9741	0.9531	0.0210

In the finetuned GeoDE rows, most models have WestAsia as the best region and SouthEastAsia as the worst region. The exception in the current table is ResNet, for which Europe is the best region and SouthEastAsia is the worst region. The best-region accuracy in finetuned GeoDE ranges from 0.9238 for ResNet to 0.9788 for ConvNeXt. The worst-region accuracy ranges from 0.8972 for ResNet to 0.9559 for ConvNeXt. This suggests that finetuned GeoDE performance is high across regions, even though a measurable regional gap remains.

The pretrained rows show more varied best and worst regions. Because pretrained object accuracy is very low, these regional rankings should be treated as unstable descriptive outputs rather than strong evidence of regional robustness. TODO: evidence needed to assess whether per-region sample sizes make the pretrained best/worst rankings reliable.

8.3 Regional Bias Gap

The regional bias gap is lowest in several pretrained rows, but those rows also have very low object accuracy. The smallest gap in the entire normalized table is 0.0039 for pretrained DeiT on GeoDE, where object accuracy is 0.0177. Other low-gap pretrained rows include pretrained SigLIP on Dollar Street with gap 0.0051 and object accuracy 0.0165, pretrained DeiT on Dollar Street with gap 0.0057 and object accuracy 0.0190, and pretrained CLIP on Dollar Street with gap 0.0057 and object accuracy 0.0146. These results indicate why object accuracy and regional gap should be

Table 4: Pretrained versus finetuned comparison

dataset	model	object acc pretrained	object acc finetuned	object acc delta	region bias gap pretrained	region bias gap finetuned	region bias gap delta
dollarstreet	clip	0.0146	0.7305	0.7159	0.0057	0.1007	0.0951
dollarstreet	convnext	0.0160	0.7667	0.7507	0.0127	0.0848	0.0720
dollarstreet	deit	0.0190	0.7512	0.7321	0.0057	0.0891	0.0835
dollarstreet	dino	0.0107	0.7340	0.7233	0.0110	0.0773	0.0663
dollarstreet	mae	0.0260	0.6674	0.6414	0.0073	0.0688	0.0614
dollarstreet	maxvit	0.0288	0.7286	0.6999	0.0254	0.1103	0.0849
dollarstreet	resnet	0.0225	0.5887	0.5662	0.0064	0.1220	0.1156
dollarstreet	siglip	0.0165	0.7621	0.7456	0.0051	0.0743	0.0692
dollarstreet	swin	0.0186	0.7609	0.7423	0.0063	0.0725	0.0663
geode	clip	0.0451	0.9538	0.9087	0.0165	0.0306	0.0141
geode	convnext	0.0239	0.9661	0.9422	0.0107	0.0229	0.0122
geode	deit	0.0177	0.9628	0.9451	0.0039	0.0200	0.0160
geode	dino	0.0210	0.9595	0.9385	0.0108	0.0238	0.0129
geode	mae	0.0329	0.9286	0.8957	0.0180	0.0187	0.0007
geode	maxvit	0.0253	0.9565	0.9312	0.0119	0.0192	0.0072
geode	resnet	0.0282	0.9086	0.8803	0.0138	0.0266	0.0128
geode	siglip	0.0252	0.9591	0.9340	0.0113	0.0278	0.0165
geode	swin	0.0203	0.9625	0.9422	0.0067	0.0210	0.0143

read together.

Among finetuned runs, GeoDE has lower regional gaps than Dollar Street. The lowest finetuned GeoDE gap is 0.0187 for MAE, followed by MaxViT at 0.0192 and DeiT at 0.0200. The highest finetuned GeoDE gap is 0.0306 for CLIP. On Dollar Street, the lowest finetuned gap is 0.0688 for MAE, followed by Swin at 0.0725 and SigLIP at 0.0743. The highest finetuned Dollar Street gap is 0.1220 for ResNet, followed by MaxViT at 0.1103 and CLIP at 0.1007.

The dataset-level contrast is visible in Table 1 and Figure 2. The mean finetuned regional gap is 0.0889 on Dollar Street and 0.0234 on GeoDE. This may indicate that, in the available evaluations, Dollar Street exposes larger regional disparities after finetuning than GeoDE. TODO: evidence needed to connect this pattern to dataset design, class distribution, region definitions, or image collection process.

8.4 Pretrained versus Finetuned Comparison

Finetuning is associated with higher object accuracy for every model on both datasets. It is also associated with a larger regional bias gap for every model on both datasets in the derived comparison table. On Dollar Street, the gap increase ranges from 0.0614 for MAE to 0.1156 for ResNet. On GeoDE, the gap increase ranges from 0.0007 for MAE to 0.0165 for SigLIP. The comparison is shown in Table 4 and Figures 3 and 4.

This pattern should not be interpreted as evidence that finetuning causes bias. A more cautious interpretation is that, in these completed evaluations, the setting with much higher object accuracy also reveals or produces larger differences between best-region and worst-region accuracy. The scatter plot in Figure 5 is useful for this point: higher object accuracy is not automatically associated with lower regional gap. In fact, several finetuned Dollar Street models have relatively high object accuracy and relatively high regional gap at the same time.

8.5 GeoDE versus Dollar Street

GeoDE and Dollar Street show different patterns. GeoDE finetuned runs have higher object accuracy and lower regional bias gaps than Dollar Street finetuned runs. Dollar Street finetuned runs have lower object accuracy and larger gaps, and the best/worst region ordering is consistent across all nine models. This may indicate that Dollar Street is a more challenging setting for the evaluated finetuned models, at least under the metrics used here. However, without verified dataset

Table 5: Regional bias gap ranking

dataset	model	finetune	loss	object acc	best region	worst region	best region acc	worst region acc	region bias gap
geode	deit	no	4.0184	0.0177	SouthEastAsia	Americas	0.0193	0.0154	0.0039
dollarstreet	siglip	no	4.2136	0.0165	as	eu	0.0182	0.0131	0.0051
dollarstreet	deit	no	4.4183	0.0190	as	am	0.0208	0.0152	0.0057
dollarstreet	clip	no	4.4294	0.0146	eu	af	0.0175	0.0118	0.0057
dollarstreet	swin	no	4.2285	0.0186	af	as	0.0224	0.0162	0.0063
dollarstreet	resnet	no	4.1654	0.0225	eu	am	0.0262	0.0199	0.0064
geode	swin	no	3.8006	0.0203	Europe	Africa	0.0238	0.0171	0.0067
dollarstreet	mae	no	4.1876	0.0260	am	eu	0.0292	0.0219	0.0073
geode	convnext	no	3.8047	0.0239	Americas	EastAsia	0.0287	0.0180	0.0107
geode	dino	no	4.0929	0.0210	SouthEastAsia	WestAsia	0.0262	0.0153	0.0108
dollarstreet	dino	no	4.6843	0.0107	af	eu	0.0153	0.0044	0.0110
geode	siglip	no	3.7253	0.0252	WestAsia	Europe	0.0312	0.0199	0.0113
geode	maxvit	no	3.6987	0.0253	WestAsia	EastAsia	0.0312	0.0193	0.0119
dollarstreet	convnext	no	4.2868	0.0160	am	af	0.0234	0.0106	0.0127
geode	resnet	no	3.6951	0.0282	Americas	Africa	0.0351	0.0213	0.0138
geode	clip	no	3.8468	0.0451	EastAsia	WestAsia	0.0498	0.0333	0.0165
geode	mae	no	3.7293	0.0329	SouthEastAsia	Africa	0.0437	0.0257	0.0180
geode	mae	yes	0.8839	0.9286	WestAsia	SouthEastAsia	0.9392	0.9205	0.0187
geode	maxvit	yes	0.8160	0.9565	WestAsia	SouthEastAsia	0.9662	0.9470	0.0192
geode	deit	yes	0.7943	0.9628	WestAsia	SouthEastAsia	0.9720	0.9520	0.0200
geode	swin	yes	0.7906	0.9625	WestAsia	SouthEastAsia	0.9741	0.9531	0.0210
geode	convnext	yes	0.7821	0.9661	WestAsia	SouthEastAsia	0.9788	0.9559	0.0229
geode	dino	yes	0.7951	0.9595	WestAsia	SouthEastAsia	0.9714	0.9477	0.0238
dollarstreet	maxvit	no	4.1550	0.0288	eu	as	0.0452	0.0198	0.0254
geode	resnet	yes	0.9538	0.9086	Europe	SouthEastAsia	0.9238	0.8972	0.0266
geode	siglip	yes	0.7972	0.9591	WestAsia	SouthEastAsia	0.9730	0.9452	0.0278
geode	clip	yes	0.8170	0.9538	WestAsia	SouthEastAsia	0.9683	0.9377	0.0306
dollarstreet	mae	yes	1.7323	0.6674	eu	af	0.6910	0.6222	0.0688
dollarstreet	swin	yes	1.4694	0.7609	eu	af	0.7915	0.7190	0.0725
dollarstreet	siglip	yes	1.4584	0.7621	eu	af	0.7945	0.7202	0.0743
dollarstreet	dino	yes	1.5633	0.7340	eu	af	0.7668	0.6895	0.0773
dollarstreet	convnext	yes	1.4279	0.7667	eu	af	0.8061	0.7214	0.0848
dollarstreet	deit	yes	1.5178	0.7512	eu	af	0.7857	0.6966	0.0891
dollarstreet	clip	yes	1.5627	0.7305	eu	af	0.7784	0.6777	0.1007
dollarstreet	maxvit	yes	1.5243	0.7286	eu	af	0.7726	0.6623	0.1103
dollarstreet	resnet	yes	2.0059	0.5887	eu	af	0.6356	0.5136	0.1220

statistics, region sample sizes, and object-category distributions, the draft should not claim why this difference occurs.

9 Analysis and Discussion

The results suggest a trade-off-like pattern, but the draft should describe it cautiously. Finetuning is associated with much higher object accuracy for every model-dataset pair, and it is also associated with a larger regional bias gap for every model-dataset pair. One possible explanation is that the pretrained evaluation has uniformly weak performance, which compresses regional differences because most predictions are incorrect. After finetuning, models become more capable of recognizing the target classes, and differences between regions become more visible. This explanation is plausible, but it cannot establish causality and requires more evidence about training dynamics, class distributions, and per-region sample sizes.

The Dollar Street results are especially important for the main research question. Every finetuned Dollar Street model has Europe as the best region and Africa as the worst region. This is consistent with prior work suggesting that object recognition can perform unevenly across countries and income contexts in Dollar Street-like imagery [DeVries et al., 2019]. However, the current paper should not claim the same mechanism or the same demographic interpretation without further evidence. The observed pattern may reflect dataset composition, object categories, image conditions,

Table 6: Best and worst region metrics

dataset	model	finetune	best region	best region acc	worst region	worst region acc	region bias gap
dollarstreet	clip	no	eu	0.0175	af	0.0118	0.0057
dollarstreet	clip	yes	eu	0.7784	af	0.6777	0.1007
dollarstreet	convnext	no	am	0.0234	af	0.0106	0.0127
dollarstreet	convnext	yes	eu	0.8061	af	0.7214	0.0848
dollarstreet	deit	no	as	0.0208	am	0.0152	0.0057
dollarstreet	deit	yes	eu	0.7857	af	0.6966	0.0891
dollarstreet	dino	no	af	0.0153	eu	0.0044	0.0110
dollarstreet	dino	yes	eu	0.7668	af	0.6895	0.0773
dollarstreet	mae	no	am	0.0292	eu	0.0219	0.0073
dollarstreet	mae	yes	eu	0.6910	af	0.6222	0.0688
dollarstreet	maxvit	no	eu	0.0452	as	0.0198	0.0254
dollarstreet	maxvit	yes	eu	0.7726	af	0.6623	0.1103
dollarstreet	resnet	no	eu	0.0262	am	0.0199	0.0064
dollarstreet	resnet	yes	eu	0.6356	af	0.5136	0.1220
dollarstreet	siglip	no	as	0.0182	eu	0.0131	0.0051
dollarstreet	siglip	yes	eu	0.7945	af	0.7202	0.0743
dollarstreet	swin	no	af	0.0224	as	0.0162	0.0063
dollarstreet	swin	yes	eu	0.7915	af	0.7190	0.0725
geode	clip	no	EastAsia	0.0498	WestAsia	0.0333	0.0165
geode	clip	yes	WestAsia	0.9683	SouthEastAsia	0.9377	0.0306
geode	convnext	no	Americas	0.0287	EastAsia	0.0180	0.0107
geode	convnext	yes	WestAsia	0.9788	SouthEastAsia	0.9559	0.0229
geode	deit	no	SouthEastAsia	0.0193	Americas	0.0154	0.0039
geode	deit	yes	WestAsia	0.9720	SouthEastAsia	0.9520	0.0200
geode	dino	no	SouthEastAsia	0.0262	WestAsia	0.0153	0.0108
geode	dino	yes	WestAsia	0.9714	SouthEastAsia	0.9477	0.0238
geode	mae	no	SouthEastAsia	0.0437	Africa	0.0257	0.0180
geode	mae	yes	WestAsia	0.9392	SouthEastAsia	0.9205	0.0187
geode	maxvit	no	WestAsia	0.0312	EastAsia	0.0193	0.0119
geode	maxvit	yes	WestAsia	0.9662	SouthEastAsia	0.9470	0.0192
geode	resnet	no	Americas	0.0351	Africa	0.0213	0.0138
geode	resnet	yes	Europe	0.9238	SouthEastAsia	0.8972	0.0266
geode	siglip	no	WestAsia	0.0312	Europe	0.0199	0.0113
geode	siglip	yes	WestAsia	0.9730	SouthEastAsia	0.9452	0.0278
geode	swin	no	Europe	0.0238	Africa	0.0171	0.0067
geode	swin	yes	WestAsia	0.9741	SouthEastAsia	0.9531	0.0210

socioeconomic context, model pretraining data, or interactions among these factors.

GeoDE shows a different pattern. Finetuned object accuracies are high across all nine models, and regional gaps are smaller than on Dollar Street. Most finetuned GeoDE models have WestAsia as the best region and SouthEastAsia as the worst region. This suggests that GeoDE still reveals regional variation even when overall performance is high. At the same time, the lower gap values indicate that the regional disparities measured by this paper's metric are smaller on GeoDE than on Dollar Street. TODO: evidence needed to explain whether GeoDE's dataset construction or class distribution is associated with this smaller measured gap.

High object accuracy is not always associated with low regional bias gap. ConvNeXt has the highest finetuned object accuracy on both datasets: 0.7667 on Dollar Street and 0.9661 on GeoDE. Its finetuned regional gap is 0.0848 on Dollar Street and 0.0229 on GeoDE. MAE has lower finetuned object accuracy than ConvNeXt on both datasets, but it has the lowest finetuned gap on both datasets: 0.0688 on Dollar Street and 0.0187 on GeoDE. This suggests that accuracy and regional stability are not the same empirical property. This is consistent with worst-group evaluation arguments that aggregate accuracy can obscure group-specific performance differences

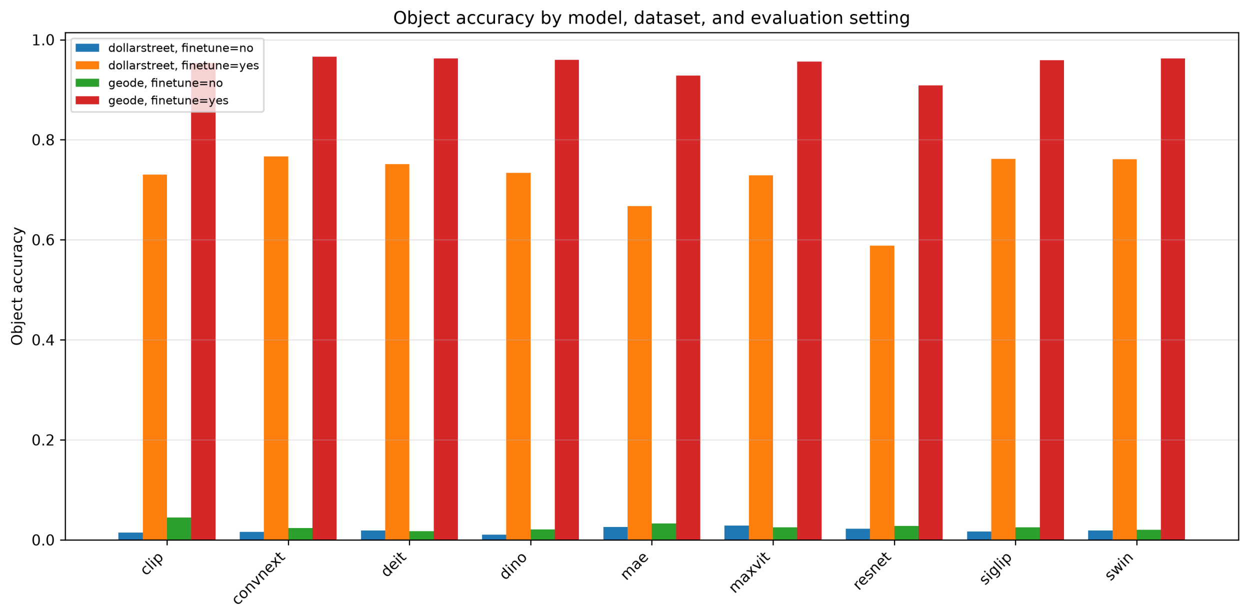

Figure 1: Object accuracy by model, dataset, and evaluation setting.

[Sagawa et al., 2020].

The model-family comparison is suggestive but incomplete. In a rough grouping, ConvNeXt and ResNet represent CNN models; DeiT, Swin, and MaxViT represent transformer models; DINO and MAE represent self-supervised models; and CLIP and SigLIP represent vision-language models. Averaging from the normalized table, the finetuned transformer group has mean object accuracy 0.8537 and mean regional gap 0.0554, while the finetuned vision-language group has mean object accuracy 0.8514 and mean regional gap 0.0584. The self-supervised group has lower mean finetuned object accuracy at 0.8224 but also the lowest mean finetuned regional gap at 0.0471. The CNN group has mean finetuned object accuracy 0.8075 and mean finetuned regional gap 0.0641. These family-level numbers are useful as a first draft summary, but a dedicated family-level table should be added before making strong claims.

The vision-language results are mixed. CLIP has high finetuned GeoDE accuracy but the largest finetuned GeoDE gap, while SigLIP is among the stronger Dollar Street finetuned models by object accuracy and has a lower Dollar Street gap than CLIP. Prior CLIP robustness work suggests that properties of pretraining data may be associated with distributional robustness [Fang et al., 2022], but this paper cannot test that mechanism directly. The safest interpretation is that vision-language models in this benchmark do not uniformly dominate on both object accuracy and regional gap.

The self-supervised results are also mixed. MAE has the lowest finetuned regional gap on both datasets but lower object accuracy than several alternatives, especially on Dollar Street. DINO has higher Dollar Street finetuned object accuracy than MAE but a larger Dollar Street regional gap. DINOv2 and MAE are both commonly framed as self-supervised representation-learning approaches [Oquab et al., 2024, He et al., 2022], but the current results do not support a broad claim that self-supervision automatically improves regional robustness.

The gap increase after finetuning should be interpreted carefully. It may indicate that better class discrimination exposes regional differences; it may reflect uneven learning across regions; or it may be partly a mathematical consequence of moving from near-zero accuracy to high accuracy.

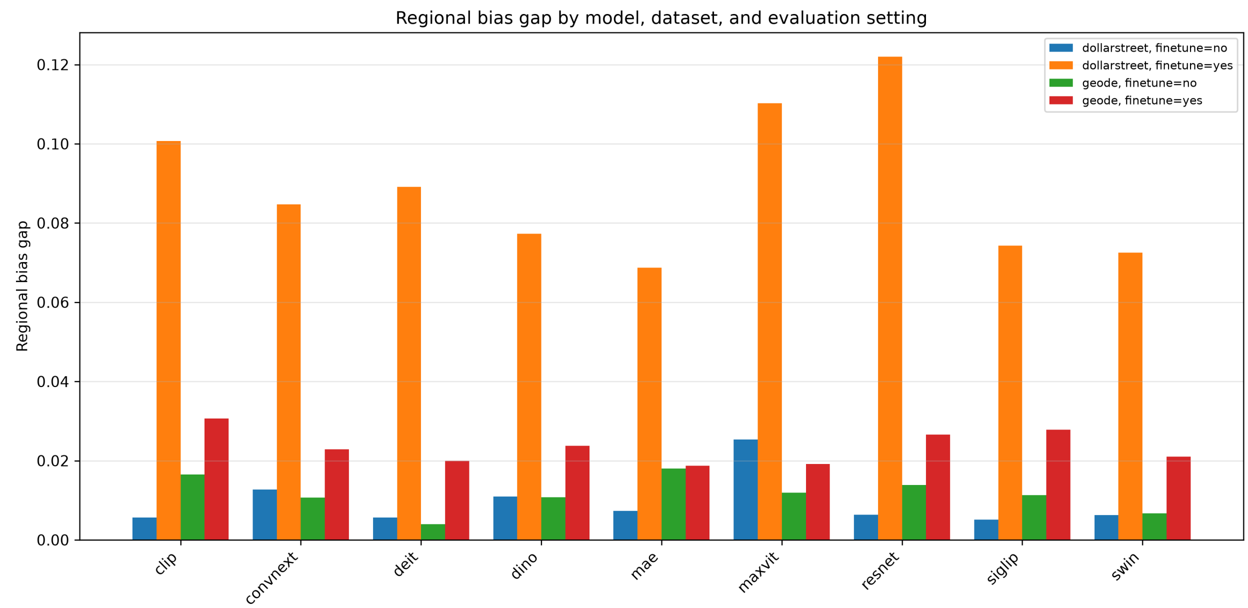

Figure 2: Regional bias gap by model, dataset, and evaluation setting.

These possibilities are not mutually exclusive. Additional analyses, such as per-class regional gaps, region sample sizes, confidence calibration, and training curves, would help distinguish them. TODO: evidence needed for per-class regional disparity and uncertainty estimates.

Overall, the results support the paper's main premise: aggregate object accuracy is not enough to characterize regional model behavior. The same model can look strong by object accuracy and still show a sizable best-worst regional gap. Conversely, a low regional gap in a very low-accuracy setting should not be interpreted as robust performance. This reinforces the need to report object accuracy, worst-region accuracy, best-region accuracy, and regional bias gap together.

10 Limitations

This draft has several important limitations. First, the analysis is based only on completed pre-trained ImageNet and finetuned result files. It does not run new training jobs, new ablations, or controlled interventions. Therefore, the paper can describe associations between evaluation setting, object accuracy, and regional bias gap, but it cannot establish causality.

Second, the regional bias gap is a simple best-minus-worst metric. It is useful because it foregrounds worst-region performance, but it does not capture all possible forms of regional fairness or robustness. It ignores uncertainty, sample size, per-class imbalance, calibration, and the distribution of errors within each region. A small gap can also be misleading when object accuracy is uniformly low.

Third, the dataset description is incomplete in this V0 draft. The paper currently avoids local dataset-statistics claims because a verified local audit table has not yet been created. The final paper should document local split sizes, region counts, class counts, and any filtering or preprocessing steps. TODO: evidence needed for local dataset statistics.

Fourth, the model-family discussion is descriptive and coarse. CNN, transformer, self-supervised, and vision-language groupings contain models that differ in architecture, pretraining data, objective, classifier head, and finetuning details. The current evidence does not support

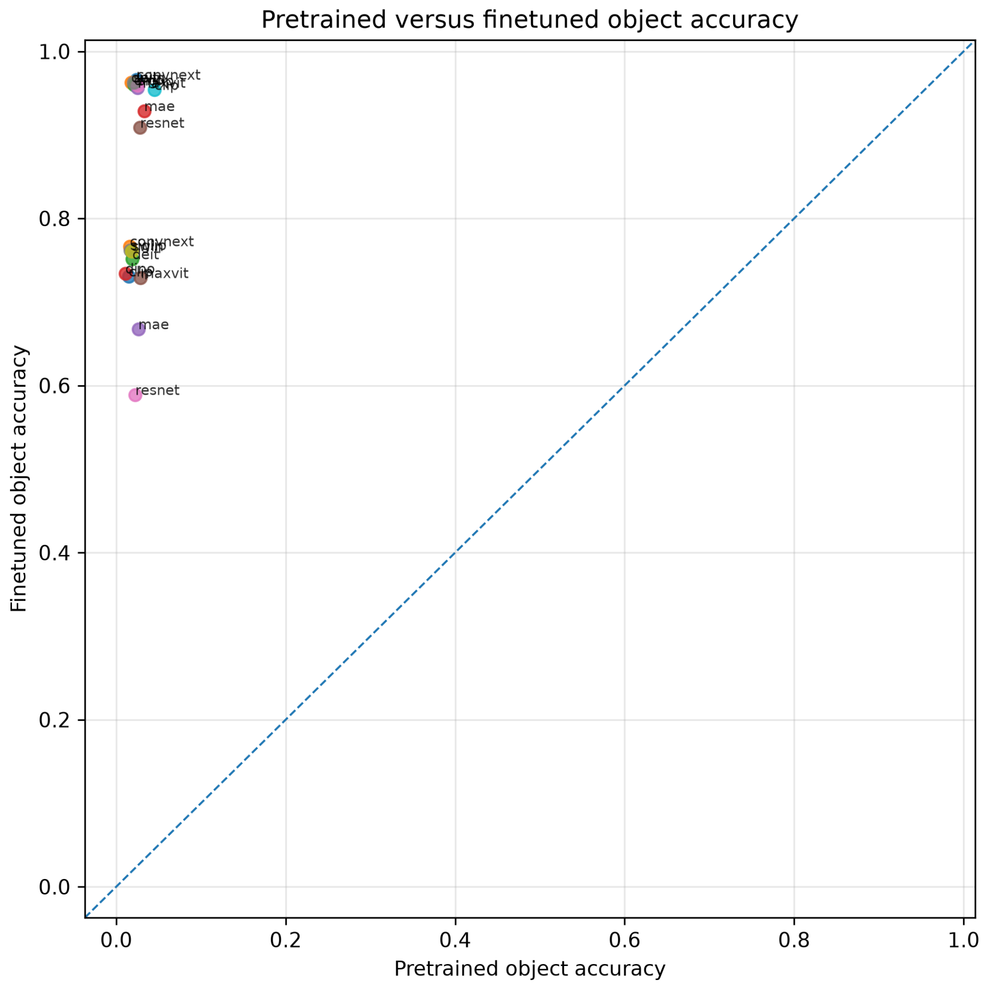

Figure 3: Pretrained versus finetuned object accuracy.

claims that a family membership causes a particular regional bias pattern.

Fifth, the pretrained ImageNet evaluation setting needs more methodological detail. The derived result files are sufficient for reporting object accuracy and regional gaps, but the final paper should more fully document classifier-head behavior, class mapping, and any model-specific evaluation differences. TODO: evidence needed for pretrained classifier-head setup and class-mapping details.

Finally, the literature review is seeded but not complete. It includes core dataset, model, and robustness papers, but it should be expanded with a transparent deep-research pass before the final paper. TODO: evidence needed for additional regional fairness and geographic vision bias literature.

11 Conclusion

This first draft analyzes regional bias in pretrained and finetuned vision models on GeoDE and Dollar Street. Using the normalized result table, the draft finds that finetuning is associated with much higher object accuracy for every model-dataset pair. It also finds that finetuning is associated

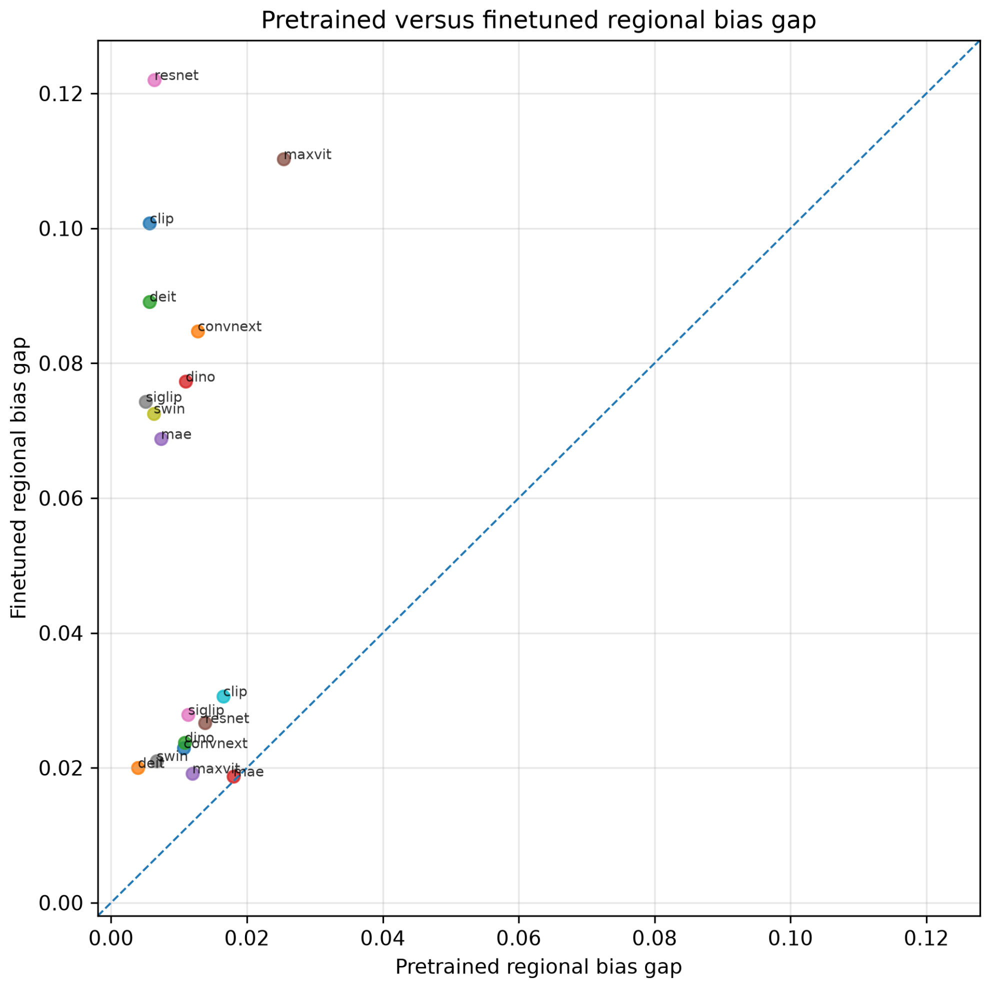

Figure 4: Pretrained versus finetuned regional bias gap.

with a larger regional bias gap for every model-dataset pair. This suggests that higher aggregate object accuracy does not automatically imply lower regional disparity.

The dataset comparison is also important. Finetuned GeoDE results show higher mean object accuracy and lower mean regional bias gap than finetuned Dollar Street results. Dollar Street finetuned results show a consistent best-region and worst-region pattern across all evaluated models, with `eu` as the best region and `af` as the worst region. GeoDE finetuned results show smaller gaps, with most models identifying WestAsia as the best region and SouthEastAsia as the worst region.

Across models, ConvNeXt has the highest finetuned object accuracy on both datasets, while MAE has the lowest finetuned regional bias gap on both datasets. This indicates that the model with the strongest object accuracy is not necessarily the model with the smallest regional gap. The paper therefore argues for reporting object accuracy, best-region accuracy, worst-region accuracy, and regional bias gap together.

These findings remain preliminary. The draft does not establish causal mechanisms, and several evidence gaps remain around local dataset statistics, pretrained classifier-head details, and model-family interpretation. The next version should expand the literature review, add a local dataset audit, add a family-level summary table, and revise the discussion after those evidence gaps are addressed.

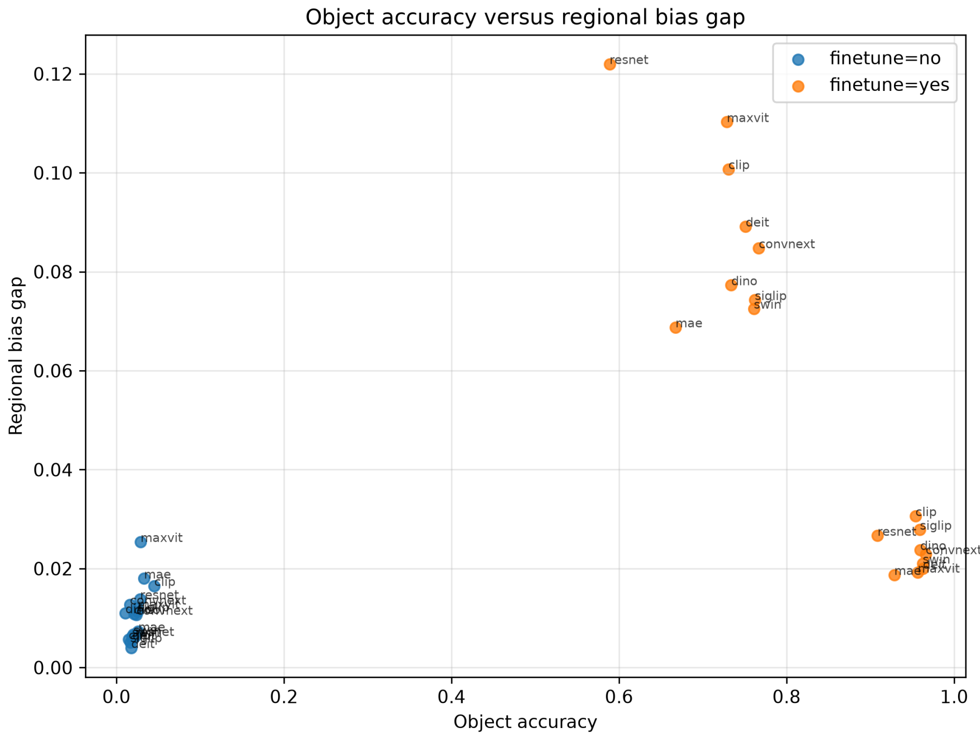

Figure 5: Object accuracy versus regional bias gap.

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